

DEVAANSH KUMAR

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

CAREER PROFILE

- UNSW Computer Science graduate and software engineer with professional experience across backend and full-stack development using Python/FastAPI, TypeScript, React/Next.js, SQL/PostgreSQL and Docker
- Experience delivering production software, network-resilience systems and applied AI/ML solutions across professional engineering roles, an industry-client capstone and independent projects

EDUCATION

Bachelor of Science (Computer Science)

September 2023 - September 2026

University of New South Wales

- UNSW International Student Excellence Award - A merit-based scholarship worth AUD 8,800
- UNSW Employability Award - 2026
- Member of CSESoc, DevSoc and CSE Revue

TECHNICAL SKILLS

- Programming: Python, TypeScript, JavaScript, SQL, Java, C, C++, Rust, R, Bash/Shell
- Frontend & Backend: React, Next.js, HTML/CSS, Node.js, FastAPI, Express, REST APIs, OpenAPI, WebSockets, Prisma, Drizzle ORM, SQLAlchemy
- Data & AI: PyTorch, Pandas, XGBoost, Machine Learning, Computer Vision, Regression & Classification
- Databases & Infrastructure: PostgreSQL, MySQL, Docker, Git, Linux, GitHub Actions, CI/CD, Alembic, Vercel
- Testing & Engineering Tools: Pytest, Vitest, Playwright, Agile/Scrum, Jira, PostHog
- CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Operating Systems, Database Systems

EXPERIENCE

Junior Software Engineer

August 2026 - September 2026

Arms Operations Analysis Pty Ltd, Sydney

- Selected for paid follow-on engineering after a five-person industry-client capstone to mature the Autonomous Network Correction and Recovery (ANCR) concept demonstrator
- Engineered Python/FastAPI and React/TypeScript capabilities for network simulation, real-time monitoring, RF/network degradation analysis, recovery workflows and operator decision support
- Integrated REST/OpenAPI services, tactical interfaces, recovery-drone simulation, Docker deployment and automated testing across multiple software prototypes

Academic Tutor

March 2026 - Present

Shalom College UNSW, Sydney

- Delivering individual and small-group tutoring across computer science, mathematics, physics, accounting and finance to 5+ recurring university students each week
- Explaining technical concepts and problem-solving approaches while providing additional academic support across a 130+ student residential community

Software Engineering Intern

October 2025 - March 2026

Tandem Learning, Sydney

- Developed and shipped 8–10 production features for a live education platform with Next.js, React, TypeScript, Prisma and Drizzle ORM within a 4–5 engineer product team
- Implemented and stabilised "Benchmark Mode", a multi-step assessment workflow spanning configuration, attempts and results, while resolving App Router, TypeScript, routing and production-build issues
- Improved the Wordsmith AI autograder, debugging prompt and grading-pipeline behaviour for visual questions to produce more reliable, evidence-grounded assessment feedback
- Refactored and extended "Agent0" AI chat with the Vercel AI SDK, working across streamed AI responses, tool integrations, prompt architecture and client/server message handling
- Integrated product analytics and resolved production issues across frontend rendering, APIs, validation, database-backed workflows and CI/build pipelines on a platform supporting 1,700+ students and 8100+ assessment attempts

Treasurer

October 2024 - February 2026

Shalom College UNSW, Sydney

- Managed a \$35,000+ student budget supporting 130+ residents across 20+ events, overseeing budgeting, expense tracking and financial allocation
- Identified \$2,000–\$4,000 in savings while collaborating with an eight-member leadership team on operations and community initiatives

RELEVANT PROJECTS**Network Analytic Tool (NAT) - Industry Client Capstone**

June 2026 - August 2026

UNSW Industry Client Capstone | 94/100 (High Distinction)

- Architected and delivered a Dockerised Python/FastAPI and React/TypeScript network-resilience platform integrating simulation, REST APIs, OpenAPI contracts and interactive tactical visualisation
- Engineered network degradation and recovery workflows spanning resilient communications, terrain-aware RF modelling, network-health analytics, stress-event simulation and recovery-drone simulation
- Led technical delivery within a five-person engineering team across shared API integration, Docker deployment, CI/CD, automated testing, client demonstrations and final handover

RuptureLab - API Resilience Testing Workbench

August 2026 - September 2026

Independent Project | github.com/dkumar315/rupture-lab

- Developed and released a full-stack API resilience workbench with Python/FastAPI, Next.js/TypeScript and PostgreSQL to run controlled baseline, fault and recovery experiments against HTTP services
- Designed a configurable reverse proxy with fault injection, Server-Sent Events monitoring, durable experiment history and resilience contracts for automated recovery verification
- Shipped with Docker Compose, Alembic migrations, GitHub Actions CI and security/dependency checks, backed by 108 backend and 53 frontend tests, 100% project-owned coverage and Playwright end-to-end validation

Computer Vision - Wheat Crop Segmentation

February 2026 - May 2026

UNSW Computer Vision Project

- Developed and evaluated four semantic image-segmentation pipelines spanning classical computer vision, graph-based ML, clustering and deep learning
- Implemented a PyTorch workflow with U-Net and pretrained ResNet50 transfer learning, data augmentation and structured evaluation
- Achieved approximately 0.91 IoU / 0.95 F1 while analysing robustness, inference trade-offs and deployment considerations

EXTRA CURRICULAR

- Harvard CS50's Introduction to Artificial Intelligence with Python, Harvard University (In Progress)
- Electronic Arts Software Engineering Virtual Experience, Forage (Jan 2025) – Proposed gameplay feature enhancements and practised software design using C++
- AWS APAC Solutions Architecture Virtual Experience, Forage (Jan 2025) – Designed scalable cloud architecture solutions with cost and performance considerations
- TechnoXian World Robotics Championship (2022) – Led a team in an international robotics competition